

UNIVERSITY OF SCIENCE & TECHNOLOGY IN ZEWAIL CITY

DEPARTMENT OF AEROSPACE ENGINEERING

SPC₃₃₇

Design and Analysis of Cold Gas Rocket Engine

Team Members

Abdelrahman MOHAMMED
202200281
Omar ELMETWALLY
202200057

Supervisors

Dr. AHMED ELTAWHEEL
Eng. ASMAA ALAAELDEEN

January 6, 2026

Abstract

A blowdown cold-gas rocket engine is designed and simulated using one-dimensional compressible-flow relations. The nozzle geometry is sized at a design chamber stagnation pressure of 100 bar with a heated-gas stagnation temperature of 2200 K and perfect expansion to ambient conditions. A time-marching tank model (rigid, well-mixed, adiabatic) is coupled to a choked nozzle to predict tank pressure/temperature histories, mass flow rate, and thrust. As a bonus extension, a constant-area Rayleigh (heat-addition) duct is included upstream of the nozzle to account for stagnation-pressure loss during heating; the combustor diameter is sized at the design point for a prescribed inlet Mach number.

Contents

1	Introduction	3
1.1	Given specifications	3
1.2	Model assumptions	3
2	Governing Theory	3
2.1	Isentropic nozzle relations	3
2.2	Adiabatic blowdown of a rigid tank	4
2.3	Bonus: Rayleigh (heat-addition) duct	4
3	Nozzle Design at the 100 bar Design Point	5
3.1	Design procedure	5
3.2	Final design parameters	5
4	Numerical Method (MATLAB Implementation)	5
4.1	Base model coupling	5
4.2	Bonus model coupling (Rayleigh + nozzle)	6
5	Bonus: Combustor Sizing	6
6	Results and Discussion	7
6.1	Tank pressure and temperature histories	7
6.2	Mass flow rate and thrust time histories	7
6.3	Interpretation	7
7	Conclusion	8

1 Introduction

Cold-gas propulsion provides a simple platform for applying gas-dynamics tools: choking, nozzle expansion, and unsteady blowdown of a pressurized tank. In this project, a pressurized air tank feeds a converging–diverging nozzle. A simplified “heater” model raises the flow stagnation temperature to a prescribed maximum of 2200 K. The nozzle is designed at a chamber stagnation pressure of 100 bar and then used in a blowdown simulation starting from 200 bar.



Figure 1: Conceptual layout of the simplified rocket engine model.

1.1 Given specifications

- Tank initial conditions: $P_{t,0} = 200$ bar, $T_{t,0} = 300$ K, $V_t = 0.5$ m³.
- Heating model: maximum stagnation temperature to nozzle inlet $T_{0,\max} = 2200$ K.
- Nozzle design point: $P_{0,d} = 100$ bar and perfect expansion $P_e = P_a$.
- Ambient pressure: $P_a = 1.013$ bar (sea level).

1.2 Model assumptions

Model assumptions

- Ideal-gas air with $R = 287$ J/(kg · K) and constant $\gamma = 1.4$.
- Quasi-one-dimensional nozzle; isentropic nozzle flow (nozzle efficiency = 1).
- Rigid tank, spatially uniform properties (well-mixed).
- Adiabatic tank (no heat transfer through tank wall).
- **Base model:** nozzle inlet stagnation temperature is fixed at $T_0 = 2200$ K.
- **Bonus model:** a constant-area Rayleigh duct accounts for stagnation-pressure loss during heating.

2 Governing Theory

2.1 Isentropic nozzle relations

At the nozzle inlet, the stagnation state (P_0 , T_0) determines the choked mass flow through the throat area A_t . For a calorically perfect gas with constant γ , the choked mass flow rate is

$$\dot{m} = A_t P_0 \sqrt{\frac{\gamma}{RT_0}} \left(\frac{2}{\gamma + 1} \right)^{\frac{\gamma+1}{2(\gamma-1)}}. \quad (1)$$

For isentropic expansion, the exit static pressure is related to the exit Mach number M_e by

$$\frac{P_e}{P_0} = \left(1 + \frac{\gamma-1}{2} M_e^2 \right)^{-\frac{\gamma}{\gamma-1}}, \quad (2)$$

and the exit-to-throat area ratio is

$$\frac{A_e}{A_t} = \frac{1}{M_e} \left[\frac{2}{\gamma + 1} \left(1 + \frac{\gamma - 1}{2} M_e^2 \right) \right]^{\frac{\gamma + 1}{2(\gamma - 1)}}. \quad (3)$$

The thrust is computed from

$$F = \dot{m} V_e + (P_e - P_a) A_e, \quad (4)$$

where $V_e = M_e a_e$ and $a_e = \sqrt{\gamma R T_e}$ with $T_e/T_0 = \left(1 + \frac{\gamma - 1}{2} M_e^2 \right)^{-1}$.

2.2 Adiabatic blowdown of a rigid tank

Tank mass is $m_t = P_t V_t / (R T_t)$. Mass conservation gives

$$\frac{dm_t}{dt} = -\dot{m}. \quad (5)$$

For a rigid, adiabatic, well-mixed tank discharging to a nozzle, the unsteady energy equation can be written as

$$\frac{d}{dt} (m_t c_v T_t) = -\dot{m} c_p T_t. \quad (6)$$

Combining (5) and (6) yields the useful differential relationship

$$\frac{dT_t}{T_t} = (\gamma - 1) \frac{dm_t}{m_t}, \quad \Rightarrow \quad T_t \propto m_t^{\gamma-1}, \quad P_t \propto m_t^\gamma. \quad (7)$$

In the numerical simulation, the tank is advanced in time with an explicit update:

$$m_{t,k+1} = m_{t,k} - \dot{m}_k \Delta t, \quad (8)$$

$$T_{t,k+1} = T_{t,k} \left(\frac{m_{t,k+1}}{m_{t,k}} \right)^{\gamma-1}, \quad (9)$$

$$P_{t,k+1} = P_{t,k} \left(\frac{m_{t,k+1}}{m_{t,k}} \right)^\gamma. \quad (10)$$

2.3 Bonus: Rayleigh (heat-addition) duct

In the bonus model, a constant-area, frictionless duct adds heat to the flow, raising stagnation temperature to $T_{0,2} = 2200$ K while causing a stagnation-pressure decrease $P_{0,2} < P_{0,1}$. The nozzle then uses $(P_{0,2}, T_{0,2})$ as inlet conditions.

Implementation note

Rayleigh-flow relations can be implemented through standard nondimensional functions (e.g., ratios to the sonic Rayleigh state, *) or by solving for the inlet Mach number and applying closed-form Rayleigh ratios to obtain $P_{0,2}/P_{0,1}$ for the imposed $T_{0,2}/T_{0,1}$. A numerical iteration may be used to couple combustor and nozzle such that the same $\dot{m}$ passes through both.

3 Nozzle Design at the 100 bar Design Point

3.1 Design procedure

At the design point, the nozzle is required to be perfectly expanded:

$$P_e = P_a, \quad P_{0,d} = 100 \text{ bar}, \quad T_{0,d} = 2200 \text{ K}. \quad (11)$$

The exit Mach number M_e is obtained from (2) using $P_e/P_{0,d} = P_a/P_{0,d}$, then the area ratio follows from (3). Finally, selecting the throat size fixes the mass flow rate from (1) and the design thrust from (4).

3.2 Final design parameters

Table 1 summarizes the final nozzle sizing results from the implemented MATLAB tool.

Table 1: Nozzle design results at $p_{0,d} = 100 \text{ bar}$ and $T_{0,d} = 2200 \text{ K}$ with $p_e = p_a$.

Quantity	Value	Unit
Throat diameter, D_t	20.0	mm
Exit Mach number, M_e	3.68	–
Area ratio, A_e/A_t	8.05	–
Exit diameter, D_e	56.7	mm
Design thrust, F_d	4865.13	N

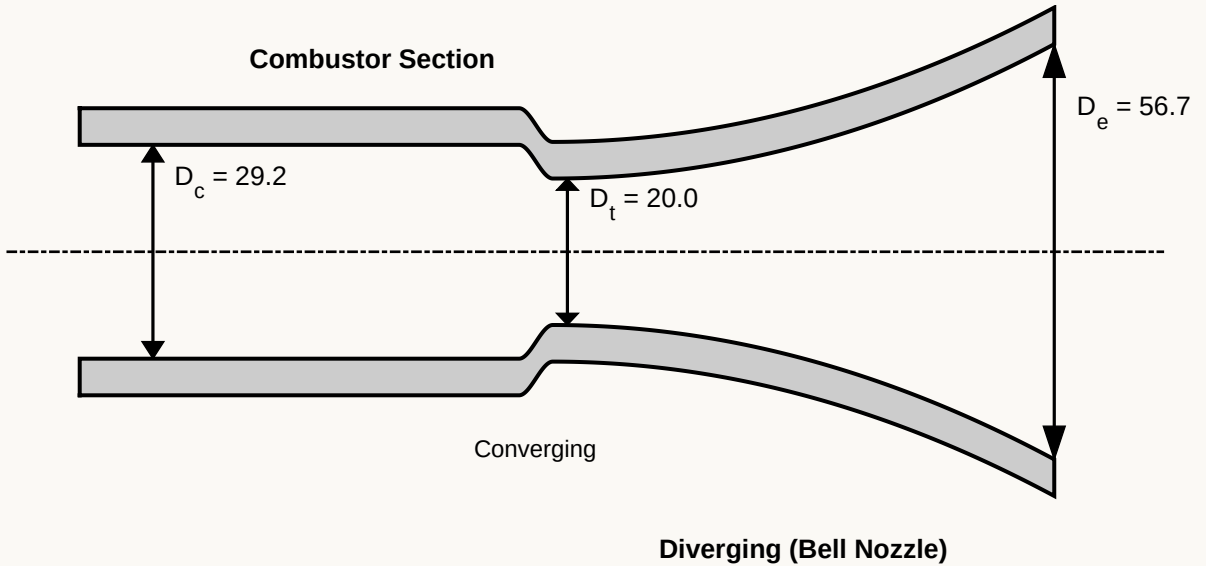


Figure 2: Nozzle geometry at the design point.

4 Numerical Method (MATLAB Implementation)

4.1 Base model coupling

At each time step t_k :

1. Set nozzle inlet stagnation pressure from the tank: $P_{0,k} = P_{t,k}$.
2. Set nozzle inlet stagnation temperature: $T_{0,k} = 2200 \text{ K}$.

3. Compute $\dot{m}_k$ using the choked relation (1).
4. Compute $P_{e,k}$, $T_{e,k}$, and $V_{e,k}$ using fixed M_e and isentropic relations.
5. Compute thrust using (4).
6. Update (m_t, T_t, P_t) using the blowdown update equations.

4.2 Bonus model coupling (Rayleigh + nozzle)

For the bonus model:

1. Use tank conditions to define combustor inlet stagnation state $(P_{0,1}, T_{0,1}) \approx (P_t, T_t)$.
2. Apply Rayleigh heat addition to reach $T_{0,2} = 2200$ K and compute the stagnation-pressure loss $P_{0,2}$.
3. Feed the nozzle with $(P_{0,2}, T_{0,2})$ and compute $\dot{m}$ and thrust.
4. Update the tank using $\dot{m}$ as in the base model.

5 Bonus: Combustor Sizing

At the design point, a low inlet Mach number was selected for the combustor:

$$M_{c,1} = 0.10. \quad (12)$$

Using the design mass flow rate and the combustor inlet state, the required constant-area duct diameter is obtained from

$$\dot{m} = \rho_1 A_c V_1, \quad D_c = \sqrt{\frac{4A_c}{\pi}}. \quad (13)$$

Table 2: Combustor sizing (bonus model) at the design point.

Quantity	Value	Unit
Combustor inlet Mach, $M_{c,1}$	0.10	—
Combustor diameter, D_c	29.2	mm

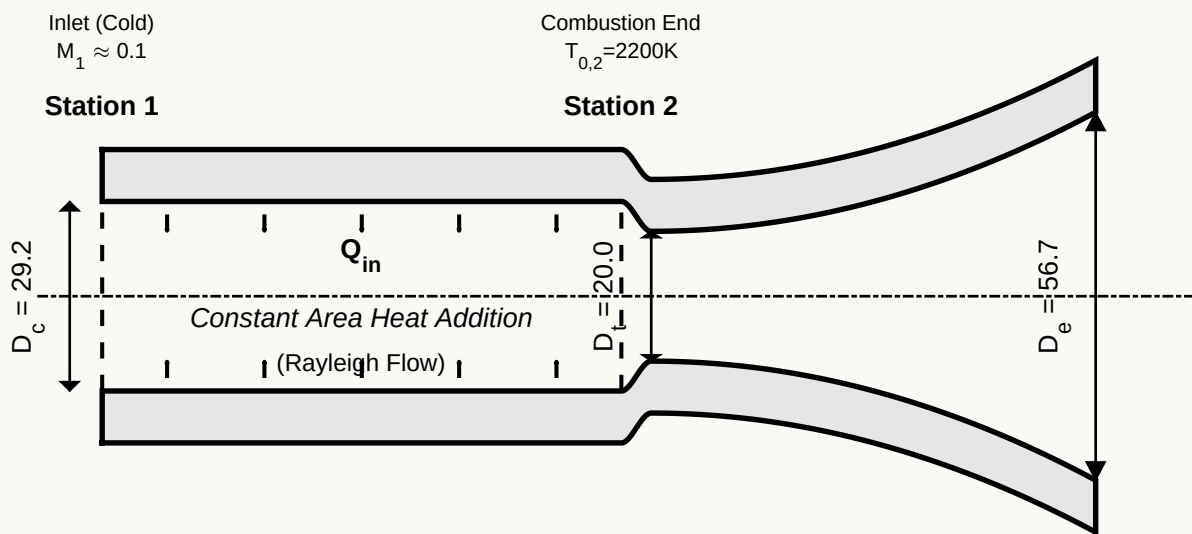


Figure 3: Bonus model: constant-area heat-addition (Rayleigh) duct upstream of the nozzle.

6 Results and Discussion

6.1 Tank pressure and temperature histories

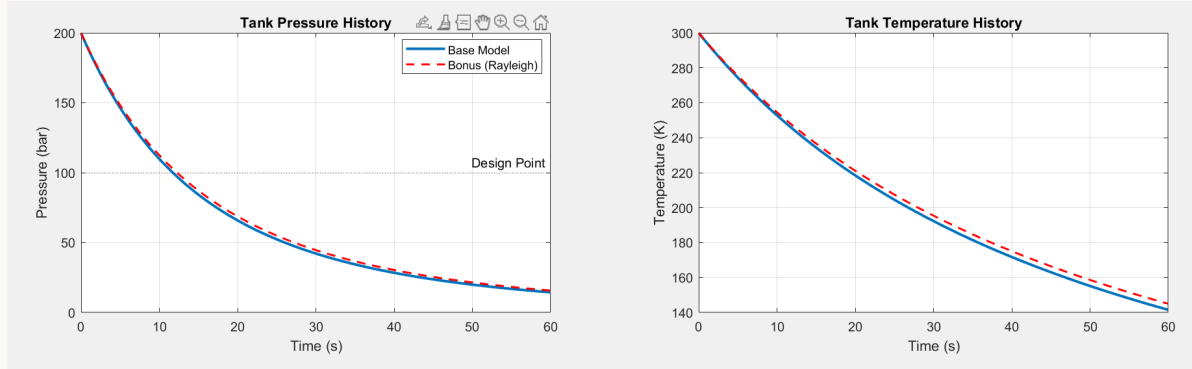


Figure 4: Tank pressure and temperature histories for base and bonus models.

6.2 Mass flow rate and thrust time histories

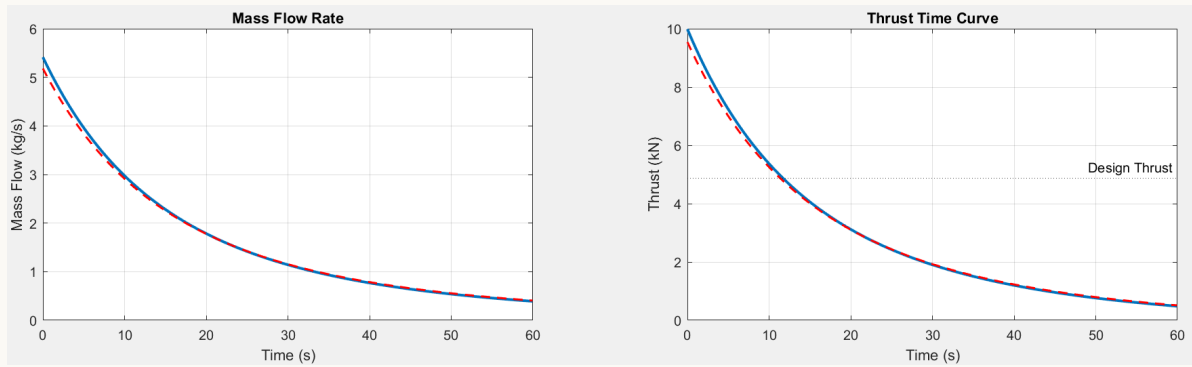


Figure 5: Mass flow rate and thrust time histories. Include both base and bonus curves for comparison.

6.3 Interpretation

- **Blowdown trend:** $P_t(t)$ and $T_t(t)$ decrease over time due to discharge from an adiabatic rigid tank.
- **Choked nozzle:** for most of the discharge, the throat remains choked and $\dot{m} \propto P_0/\sqrt{T_0}$.
- **Bonus effect:** Rayleigh heat addition reduces stagnation pressure across the duct, which lowers the effective nozzle inlet pressure and therefore reduces $\dot{m}$ and thrust relative to the base model (all else equal).

Sanity checks reported in this work

- Perfect expansion at design: $P_e \approx P_a$ at $P_0 = 100$ bar using $M_e = 3.68$.
- Geometric consistency: $D_e \approx D_t \sqrt{A_e/A_t}$.
- Thrust consistency: at design, pressure thrust term is small because $P_e \approx P_a$; thrust is dominated by $\dot{m} V_e$.

7 Conclusion

A CD nozzle was sized at a 100 bar design point for a heated-gas stagnation temperature of 2200 K and perfect expansion to ambient pressure. A coupled blowdown + choked-nozzle simulation predicts the transient tank conditions, mass flow rate, and thrust. Including a Rayleigh heat-addition duct provides a more realistic prediction by introducing stagnation-pressure losses during heating and requires sizing a combustor diameter consistent with a low inlet Mach number.

Appendix A: Key constants used

Table 3: Thermophysical constants and environment.

Quantity	Value	Unit
Specific gas constant, R	287	J/(kg·K)
Ratio of specific heats, γ	1.4	—
Ambient pressure, P_a	1.013	bar
Tank volume, V_t	0.5	m ³
Initial tank pressure, $P_{t,0}$	200	bar
Initial tank temperature, $T_{t,0}$	300	K

Appendix B: MATLAB code

Listing 1: Skeleton of the time-marching solver.

```

1 clear; clc; close all;
2 gamma = 1.4;
3 R = 287;
4 Pa = 101325;
5 P0_design = 100 * 1e5;
6 T0_combustor = 2200;
7 P_tank_init = 200 * 1e5;
8 T_tank_init = 300;
9 Vol_tank = 0.5;
10 Dt = 0.020;
11 At = pi * (Dt/2)^2;
12 fprintf('---DESIGN PARAMETERS---\n');
13 fprintf('Throat Diameter: %.1f mm\n', Dt*1000);
14 target_PR = P0_design / Pa;
15 find_Mach = @(M) (1 + (gamma-1)/2 * M.^2).^(gamma/(gamma-1)) - target_PR;
16 Me_design = fzero(find_Mach, 3.5);
17 Ae_At = (1/Me_design) * ((2/(gamma+1)) * (1 + (gamma-1)/2 * Me_design^2)).^((gamma+1)/(2*(gamma-1)));
18 Ae = At * Ae_At;
19 De = sqrt(4 * Ae / pi);
20 c_star = sqrt(R * T0_combustor) / (sqrt(gamma) * (2/(gamma+1)).^((gamma+1)/(2*(gamma-1))));
21 mdot_design = (P0_design * At) / c_star;

```

```

22 Ve_design = Me_design * sqrt(gamma * R * (T0_combustor / (1 + (gamma-1)/2 *
    Me_design^2)));
23 Thrust_design = mdot_design * Ve_design;
24 fprintf('Design_Nozzle_Exit_Mach: %.2f\n', Me_design);
25 fprintf('Design_Area_Ratio (Ae/At): %.2f\n', Ae_At);
26 fprintf('Nozzle_Exit_Diameter: %.1f mm\n', De*1000);
27 fprintf('Design_Thrust: %.2f N\n', Thrust_design);
28 fprintf('-----\n');
29 M_inlet_design = 0.1;
30 rho_cold_design = P0_design / (R * T_tank_init);
31 V_inlet = M_inlet_design * sqrt(gamma * R * T_tank_init);
32 Ac = mdot_design / (rho_cold_design * V_inlet);
33 Dc = sqrt(4 * Ac / pi);
34 fprintf('\n---BONUS: COMBUSTOR_SIZING---\n');
35 fprintf('Combustor_Inlet_Mach (Design): %.2f\n', M_inlet_design);
36 fprintf('Combustor_Diameter: %.1f mm\n', Dc*1000);
37 fprintf('-----\n');
38 dt = 0.05;
39 t_max = 60;
40 steps = t_max / dt;
41 time = 0:dt:t_max;
42 N = length(time);
43 P_tank = zeros(1, N); T_tank = zeros(1, N);
44 mdot = zeros(1, N); Thrust = zeros(1, N);
45 m_gas = zeros(1, N);
46 P_tank_B = zeros(1, N); T_tank_B = zeros(1, N);
47 mdot_B = zeros(1, N); Thrust_B = zeros(1, N);
48 P0_loss = zeros(1, N);
49 P_tank(1) = P_tank_init; T_tank(1) = T_tank_init;
50 m_gas(1) = (P_tank_init * Vol_tank) / (R * T_tank_init);
51 P_tank_B(1) = P_tank_init; T_tank_B(1) = T_tank_init;
52 m_gas_B = m_gas(1);
53 for k = 1:N-1
54
55     if P_tank(k) > 1.5 * Pa
56         P0_noz = P_tank(k);
57         mdot(k) = (P0_noz * At) / c_star;
58
59         P_exit = P0_noz / target_PR;
60         Te = T0_combustor / (1 + (gamma-1)/2 * Me_design^2);
61         Ve = Me_design * sqrt(gamma * R * Te);
62         Thrust(k) = mdot(k) * Ve + (P_exit - Pa) * Ae;
63
64         m_gas(k+1) = m_gas(k) - mdot(k) * dt;
65         T_tank(k+1) = T_tank(k) * (m_gas(k+1) / m_gas(k))^(gamma-1);
66         P_tank(k+1) = (m_gas(k+1) * R * T_tank(k+1)) / Vol_tank;
67     else
68         P_tank(k+1) = P_tank(k); T_tank(k+1) = T_tank(k);
69         mdot(k) = 0; Thrust(k) = 0;
70     end
71
72     if P_tank_B(k) > 1.5 * Pa
73

```

```

74     mdot_guess = mdot(k);
75
76     func_P0_star = @(M) ((gamma+1)/(1+gamma*M^2)) * ...
77         ((2/(gamma+1)) * (1+(gamma-1)/2*M^2))^(gamma/(gamma-1));
78
79     func_T0_star = @(M) (2*(gamma+1)*M^2 * (1 + (gamma-1)/2*M^2)) / (1 + gamma*M
80         ^2)^2;
81     for iter = 1:15
82         rho1_stag = P_tank_B(k) / (R * T_tank_B(k));
83         a1 = sqrt(gamma * R * T_tank_B(k));
84
85         M1 = (mdot_guess / (rho1_stag * Ac)) / a1;
86
87         T0_ratio_req = T0_combustor / T_tank_B(k);
88
89         val_T01_star = func_T0_star(M1);
90         val_T02_star = val_T01_star * T0_ratio_req;
91
92         if val_T02_star > 1.0
93             val_T02_star = 0.999;
94         end
95
96         solve_M2 = @(m) func_T0_star(m) - val_T02_star;
97         M2 = fzero(solve_M2, [M1, 0.9999]);
98
99         P0_ratio = func_P0_star(M2) / func_P0_star(M1);
100
101         P0_noz_B = P_tank_B(k) * P0_ratio;
102
103         mdot_new = (P0_noz_B * At) / c_star;
104         if abs(mdot_new - mdot_guess) < 1e-5
105             mdot_guess = mdot_new;
106             break;
107         end
108         mdot_guess = mdot_new;
109     end
110
111     mdot_B(k) = mdot_guess;
112     Thrust_B(k) = mdot_B(k) * Ve + ((P0_noz_B/target_PR) - Pa) * Ae;
113
114     m_curr = (P_tank_B(k) * Vol_tank) / (R * T_tank_B(k));
115     m_next = m_curr - mdot_B(k) * dt;
116     T_tank_B(k+1) = T_tank_B(k) * (m_next / m_curr)^(gamma-1);
117     P_tank_B(k+1) = (m_next * R * T_tank_B(k+1)) / Vol_tank;
118
119 else
120     P_tank_B(k+1) = P_tank_B(k); T_tank_B(k+1) = T_tank_B(k);
121     mdot_B(k) = 0; Thrust_B(k) = 0;
122 end
123
124 time = time(1:end-1);
125 P_tank = P_tank(1:end-1); T_tank = T_tank(1:end-1);
126 mdot = mdot(1:end-1); Thrust = Thrust(1:end-1);

```

```

126 P_tank_B = P_tank_B(1:end-1); T_tank_B = T_tank_B(1:end-1);
127 mdot_B = mdot_B(1:end-1); Thrust_B = Thrust_B(1:end-1);
128 figure('Name', 'Rocket_Engine_Performance', 'Position', [100, 100, 1000, 600]);
129 subplot(2,2,1);
130 plot(time, P_tank/1e5, 'LineWidth', 2); hold on;
131 plot(time, P_tank_B/1e5, '--r', 'LineWidth', 1.5);
132 yline(100, 'k:', 'Design_Point');
133 xlabel('Time_(s)'); ylabel('Pressure_(bar)');
134 title('Tank_Pressure_History');
135 legend('Base_Model', 'Bonus_(Rayleigh)', 'Location','best');
136 grid on;
137 subplot(2,2,2);
138 plot(time, T_tank, 'LineWidth', 2); hold on;
139 plot(time, T_tank_B, '--r', 'LineWidth', 1.5);
140 xlabel('Time_(s)'); ylabel('Temperature_(K)');
141 title('Tank_Temperature_History');
142 grid on;
143 subplot(2,2,3);
144 plot(time, mdot, 'LineWidth', 2); hold on;
145 plot(time, mdot_B, '--r', 'LineWidth', 1.5);
146 xlabel('Time_(s)'); ylabel('Mass_Flow_(kg/s)');
147 title('Mass_Flow_Rate');
148 grid on;
149 subplot(2,2,4);
150 plot(time, Thrust/1000, 'LineWidth', 2); hold on;
151 plot(time, Thrust_B/1000, '--r', 'LineWidth', 1.5);
152 yline(Thrust_design/1000, 'k:', 'Design_Thrust');
153 xlabel('Time_(s)'); ylabel('Thrust_(kN)');
154 title('Thrust_Time_Curve');
155 grid on;
156 fprintf('\nSimulation_Complete.\n');

```